



Programme: B E – Information Science & Engineering

Internal Assessment – II

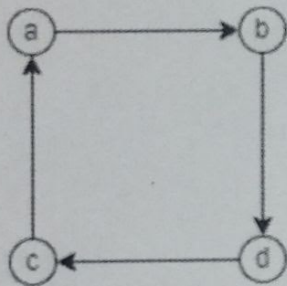
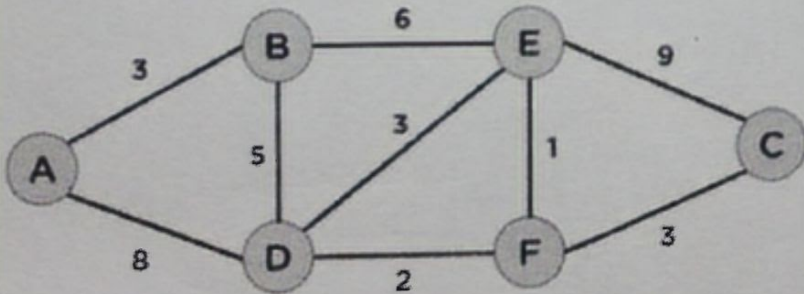
TERM : 15 April 2024-27 July 2024	COURSE NAME: Design and Analysis of Algorithms
DATE : 24-07-2024	TIME:9:30am-10:30am
COURSE CODE : IS43	
MAX MARKS: 30	Semester: 4
PORTIONS : L 17 – L42	



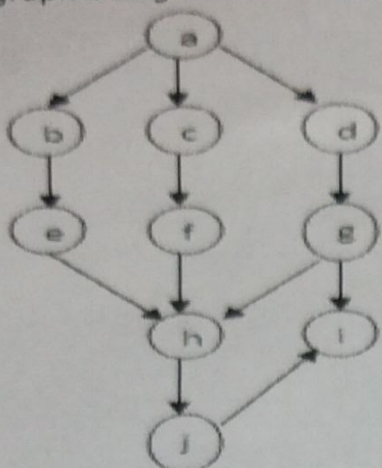
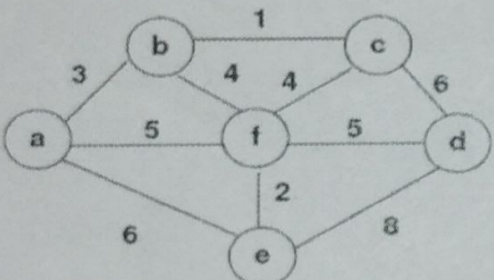
Mobile Phones are banned

Instructions to Candidates: **Answer any TWO full questions.**

Marks: **15x2=30**

Q. NO	Questions	Blooms Levels (L1 to L6)*	CO	Marks
1.a	Define a transitive closure. Obtain transitive closure for the given graph by using Warshall's algorithm. 	L3	CO3	5
b	Define AVL tree. Construct an AVL tree by successive insertion of elements 5, 6, 8, 3, 2, 4, 7.	L3	CO4	5
c	What is backtracking? Construct state space tree to solve four queen's problem using backtracking.	L3	CO5	5
2.a	Define Minimum Spanning tree. Apply Kruskal's algorithm on the following graph construct a minimum spanning tree. 	L3	CO3	5



b	Differentiate between BFS and DFS methods. Traverse the below graph using DFS method. 	L3	CO4	5
c	Discuss on various properties of heap. Use the Bottom-up construction approach to construct a heap for the list 2, 9, 7, 6, 5, 8.	L3	CO5	5
3.a	For the following graph, solve the single source shortest path problem with vertex "a" as the source using Dijkstra's approach 	L3	CO3	5
b	Design an algorithm to sort an array of elements using insertion sort and find its best and worst case time complexity.	L3	CO4	5
c	Apply heapsort for the given list 1, 12, 9, 5, 6, 10. (increasing order)	L3	CO5	5